MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, LIGHT, EXTENSION, SMALL BOATS

25 WATTS, 12-28 VOLTS, WATERTIGHT, SYMBOL 302.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for us by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

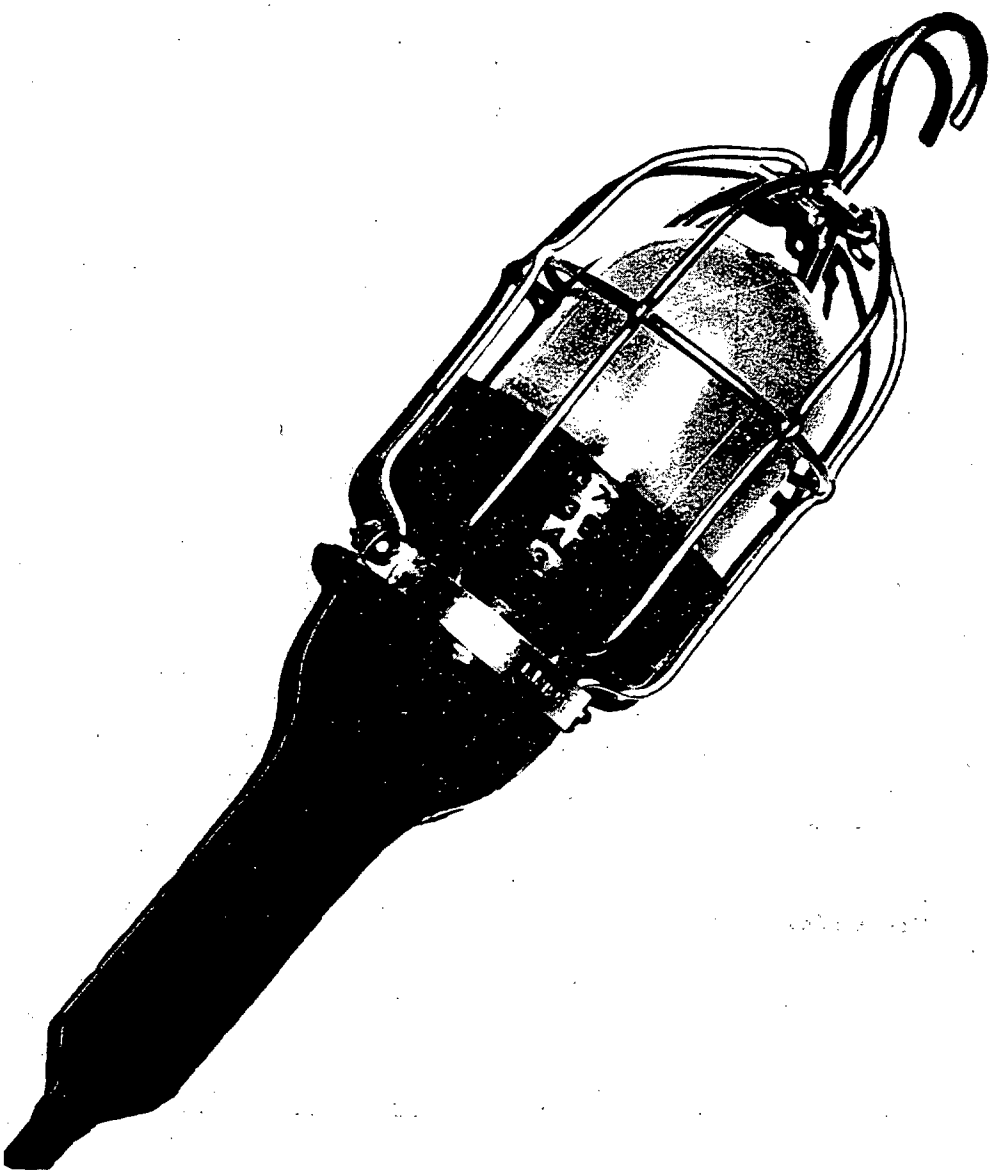


FIGURE 1. Typical view of Symbol 302.2.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

(A) denotes changes

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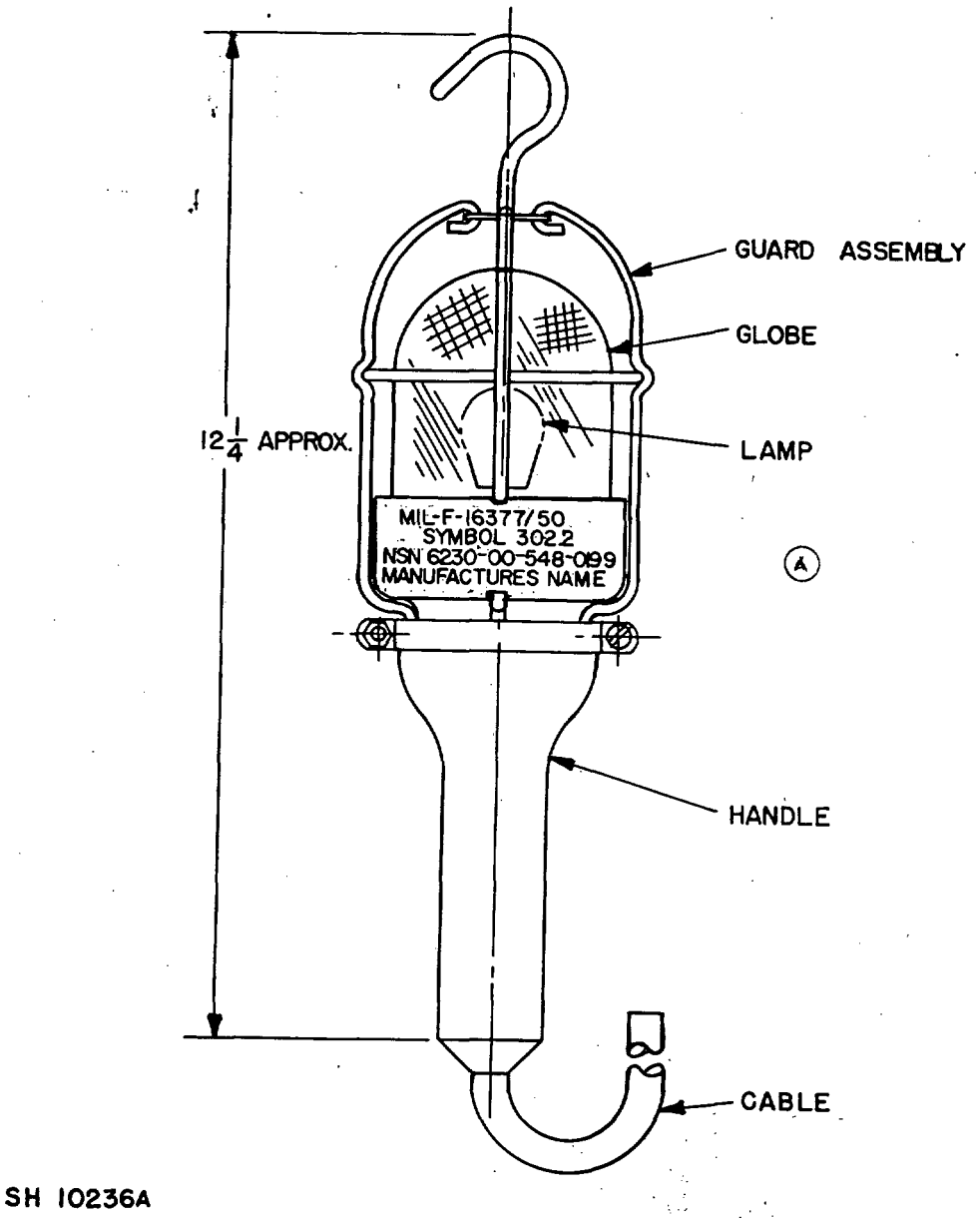


FIGURE 2. Light, extension, symbol 302.2.

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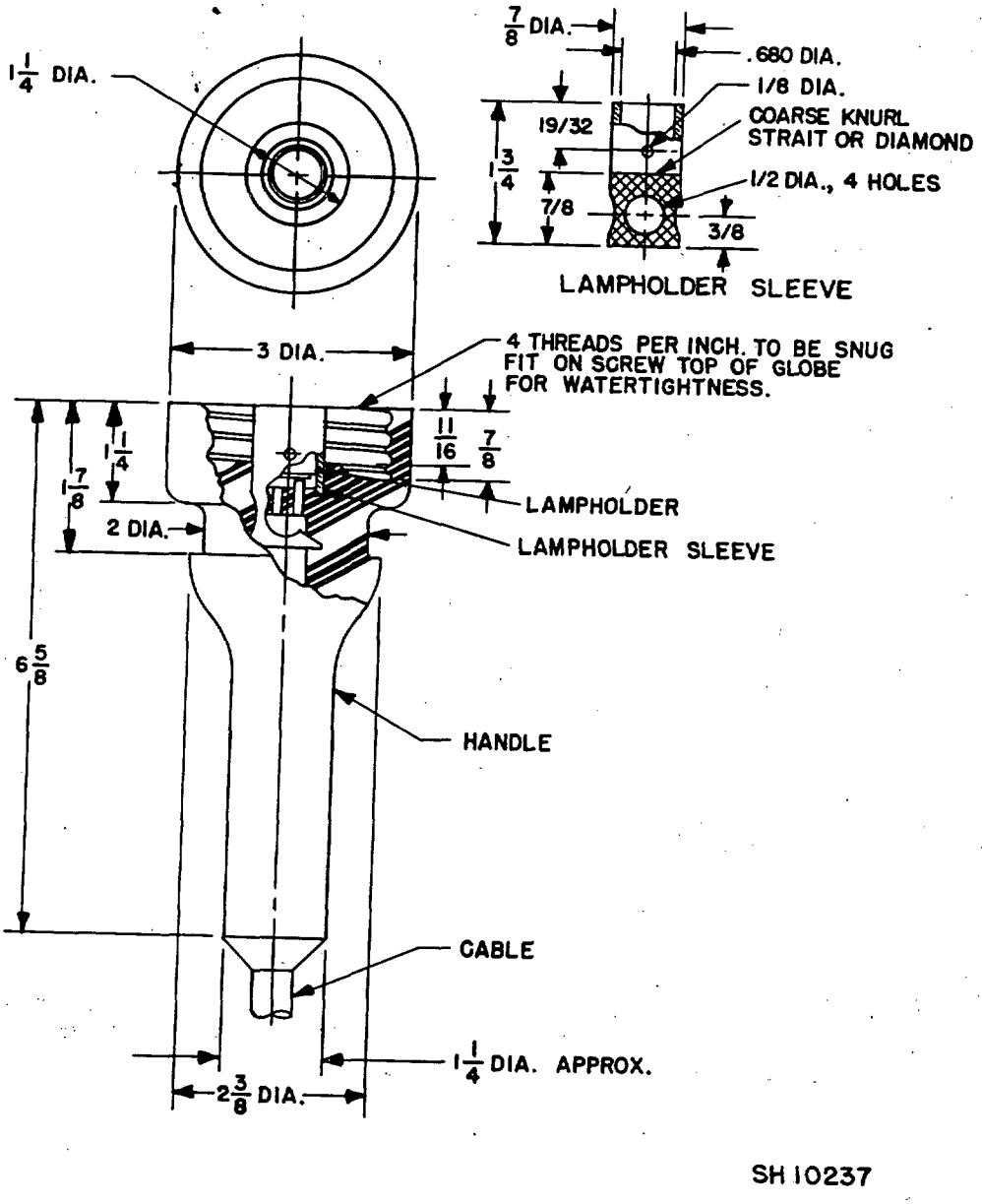
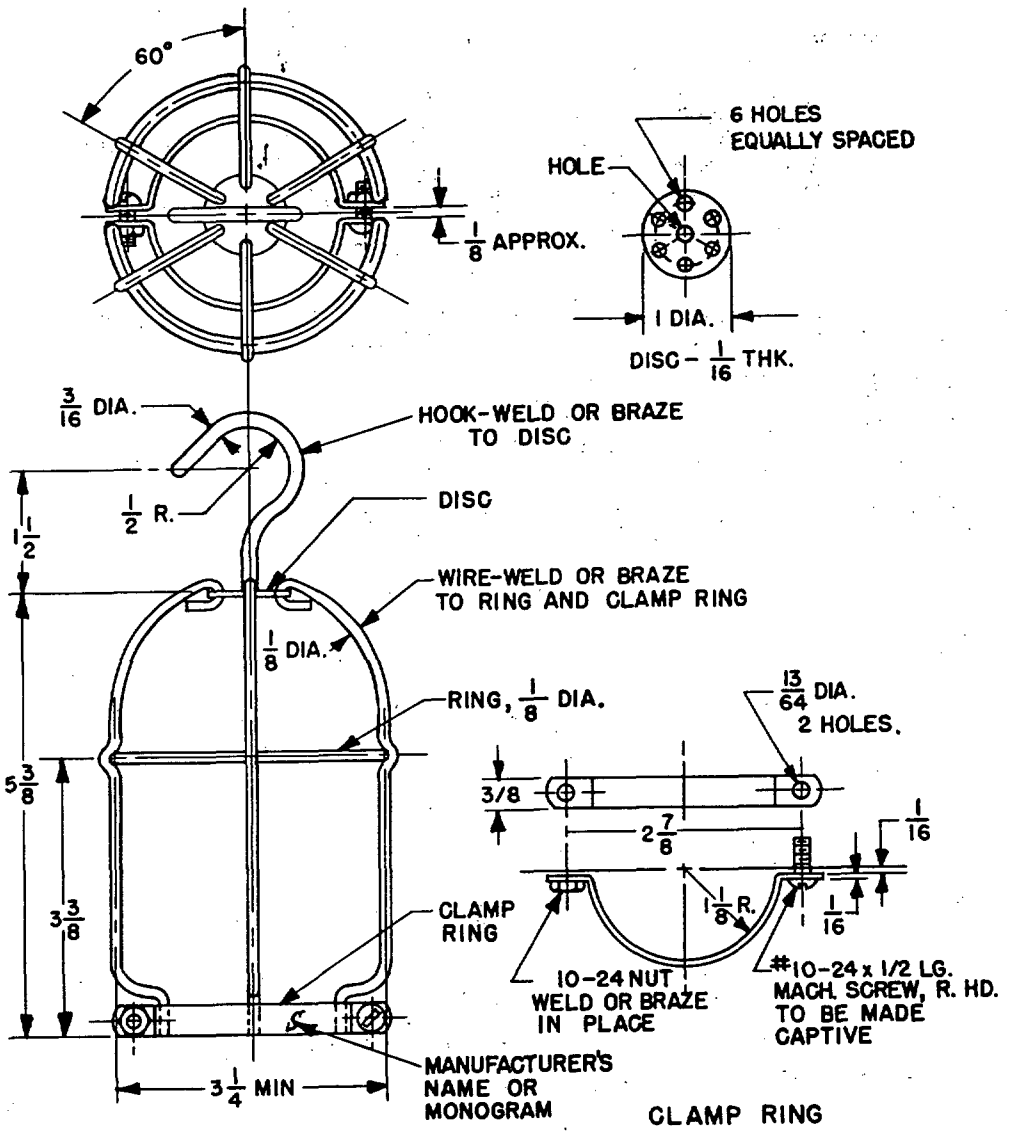


FIGURE 3. Handle assembly.

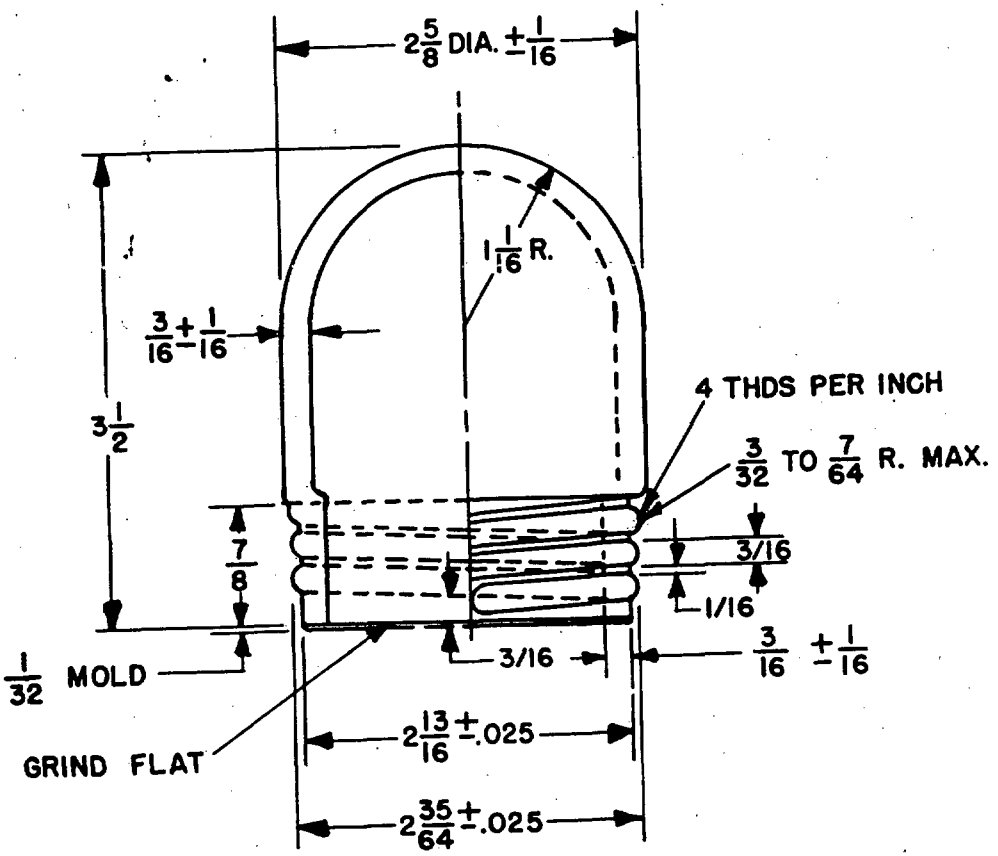
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SH 10238

FIGURE 4. Guard assembly.

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Material: High impact, heat-resistant glass.
Color: Clear transparent.
Transmittance: 85 percent minimum.

FIGURE 5. Globe.

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REQUIREMENTS:

Type: II
Class: 1
Symbol 302.2: Light, extension, portable, incandescent, 12-28 volts, 25 watts, watertight.

Design:
- Electrical rating: 12-28 volts, 25 watts maximum.
- Dimensions and configurations: See figures 1 through 5.
- Handle assembly: See figure 3.
  - Handle: The handle shall be molded neoprene rubber or polyurethane. Polyurethane, if used, must be molded in accordance with instructions furnished by the manufacturer. The handle shall be firmly molded and bonded to sheath of cable to provide a permanent, watertight seal.
  - Lampholder: Lampholder shall be in accordance with MIL-L-970/2. Contact screws shall be removed from lampholder and the conductor of cable shall be soldered in the screw holes with extreme care to insure a securely bonded electrical connection with a minimum amount of solder.
  - Lampholder sleeve: Material shall be brass.
- Cable: The light shall be furnished with 25 feet (plus 6, minus 0 inches) of cable, type SO-16/2 or ST-16/2 in accordance with J-C-580.
- Guard assembly: See figure 4. Material shall be brass or corrosion-resisting steel.
- Globe: See figure 5. A film of silicone grease shall be applied on the seating surface of globe before assembling.
- Lamp: Lamp shall be incandescent, double contact, bayonet candelabra base. Lamp is not furnished. The lamps shown in table I shall be used with the system as indicated:

TABLE I. Lamp schedule.

System	CP	Type	MS	National StockNo.	Industry number
12-16 volt	21	TS-47	15584-4	6240-00-153-6514	1142
24-28 volt	15	TS-102	15584-8	6240-00-155-8726	1692

- Enclosure effectiveness: The light shall be designed to be watertight.
- Drop: The guard and globe shall not be impaired nor shall globe threaded engagements be loosened when the light is dropped to a concrete floor ten times from a height of 10 feet. The light shall be allowed to freely fall in a manner that would normally be the case if the light were to fall from a horizontal platform to the floor.
- Weight: 5 pounds maximum.
- Identification: The information of the type I label of MIL-F-16377, shall be molded on the handle as shown on figure 2.

Quality assurance: Quality assurance shall be as specified in MIL-F-16377 and as follows:
- First article inspection:
  - Complete light: See table II.
  - Globe: See table III.
- Quality conformance:
  - Comparison inspection: Same as the first article inspection.
  - Inspection of product for delivery:
    - Examination: See MIL-F-16377
    - Examination for preparation of delivery: See MIL-F-16377
    - Quality conformance tests: See MIL-F-16377

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TABLE II. First article and comparison inspection for complete lights.

Inspection	Requirement paragraph	Inspection paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Drop	1/	1/	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Optical uniformity	3.7.5.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	4 ft./lbs.
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	
Magnetic permeability	3.4.8	4.8.18	

1/ As specified herein.

TABLE III. First article and comparison inspection for globes.

Inspection	Requirement paragraph	Inspection paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.5.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	4 ft./lbs.
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	

General information: Assignment of Military part numbers and general information is shown in table IV.

TABLE IV. Part number and general information.

Military part number M16377/50-..	Item name	National Stock Number (NSN)	Remarks
302.2	Complete light, symbol 302.2	6230-00-548-0199	1/
001	Globe	6230-00-399-6317	2/
002	Guard assembly	6230-00-215-6356	3/

1/ Supersedes MS17783 and Drawing 9000-S6401-73271.
2/ Supersedes part numbers MS17802-1 and 9000-S6405-74222-73.
3/ Supersedes part numbers MS17783-3 and 9000-S6401-73271-2.

Preparing activity:
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